

Principles of Programming Languages Paradigms: Object Oriented Programming.

Andrei Arusoaie¹

¹Department of Computer Science

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Outline

Paradigms

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Object-Oriented Programming

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Conclusion

Paradigms

- ▶ Imperative Programming: ✓

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- ▶ Interfaces, abstract classes

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- ▶ Classes establish the name of the class, the *data method* members with types and signature, visibility and implementation.
- ▶ Object = an *instance* of a class; it has a state where its *data members* (or *attributes* or *properties*) – described in the corresponding class – have values

Constructors and Destructors

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- ▶ *Access modifiers*: used to restrict/grant access to object's data

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- ▶ Single vs. multiple inheritance

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- ▶ Subtype polymorphism: uses inheritance

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 3. Methods in interfaces must be implemented by the implementing class

Recap

Concepts:

- ▶ classes, attributes/properties, methods
- ▶ access modifiers
- ▶ objects
- ▶ constructors, destructors
- ▶ encapsulation
- ▶ abstraction
- ▶ inheritance
- ▶ polymorphism (overloading, parametric, subtype)
- ▶ dynamic dispatch/late binding
- ▶ interfaces, abstract classes

Claimed advantages

- ▶ Re-usability
- ▶ Redundancy
- ▶ Easy maintenance
- ▶ Modularity
- ▶ Productivity
- ▶ Lower costs
- ▶ Higher-quality software
- ▶ Security (data hiding)

Disadvantages: what people complain about

- ▶ Size of the programs
- ▶ Effort
- ▶ Speed
- ▶ Not suitable for all types of problems
- ▶ Steep learning curve

Facts

- ▶ OOP scales; it is widely used in the industry
- ▶ Not all languages that are advertised as OOP languages support all the features above (e.g., C++ does not have GC)
- ▶ Sometimes, misunderstanding of the OOP concepts may complicate your code

Resources

- ▶ Chapter 10 in **Programming Languages: Principles and Paradigms** by Maurizio Gabbrielli and Simone Martini.
URL: <https://link.springer.com/book/10.1007/978-1-84882-914-5>